

GATE XE

Engineering Sciences

TREND-BASED MOCK PAPER

— 03 —

Duration	3 Hours
Total Questions	65
Maximum Marks	100
Sections	10 subjects

Difficulty Distribution

Easy: 20 questions (31%)

Moderate: 31 questions (48%)

Difficult: 14 questions (22%)

GENERAL INSTRUCTIONS

GATE XE — Trend-Based Mock Paper

1. Examination Pattern

The paper contains 65 questions carrying a total of 100 marks. The duration of the examination is 3 hours.

2. Question Types

Questions are of three types:

- (a) Multiple Choice Questions (MCQ): One correct answer out of four options.
 - (b) Multiple Select Questions (MSQ): One or more correct answers out of four options. Each correct answer carries full marks. There is no negative marking.
 - (c) Numerical Answer Type (NAT): A real number answer to be entered. No options are provided.
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3. Marking Scheme

- 1-mark questions: +1 for correct, -0.33 for incorrect (MCQ only)
 - 2-mark questions: +2 for correct, -0.66 for incorrect (MCQ only)
 - 3-mark questions: +3 for correct, -0.66 for incorrect (MCQ only)
 - NAT questions carry no negative marking.
 - MSQ questions carry no negative marking.
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4. How to Answer

- For MCQ: Select exactly one option.
 - For MSQ: Select one or more options. Partial credit is NOT awarded.
 - For NAT: Enter the numerical answer in the provided space. Round off as specified.
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5. Rough Work

All rough work should be done only in the space provided in this question paper. Do not use additional sheets.

6. Calculator

An on-screen virtual calculator is available for use during the examination. Physical calculators are NOT permitted.

7. Important Notes

- Do not open the seal of the Question Booklet until you are instructed to do so.
 - The Question Booklet contains multiple pages. Verify the page count.
 - Immediately report any discrepancy in the Question Booklet to the invigilator.
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Q1. 1M [-0.33]

[Basic Electrical]

V =:

- (A) IR
- (B) I/R
- (C) R/I
- (D) I^2R

Q2. 1M [no -ve]

[Environmental Science]

PM2.5 particle size:

Enter your answer here

Q3. 1M [-0.33]

[Engineering Mathematics]

If A is 2x2 with trace 5 and det 6, eigenvalues:

- (A) 2 and 3
- (B) 1 and 4
- (C) 3 and 3
- (D) 2 and 4

Q4. 1M [no -ve]

[Solid Mechanics]

Tensile stress formula:

Enter your answer here

Q5. 1M [-0.33]

[Environmental Science]

PM2.5 particle size:

- (A) $<2.5; \mu\text{m}$
- (B) $<5; \mu\text{m}$
- (C) $<10; \mu\text{m}$
- (D) $<1; \mu\text{m}$

Q6. 1M [-0.33]

[Basic Electronics]

Diode used as:

- (A) Amplifier
- (B) Rectifier
- (C) Oscillator
- (D) None

Q7. 1M [no -ve]

[Computers and Programming]

Binary search complexity:

Enter your answer here

Q8.

1M [no -ve]

[Basic Electrical]

V =:

*Enter your answer here***Q9.**

1M [no -ve]

[Computers and Programming]

Binary search complexity:

*Enter your answer here***Q10.**

1M [-0.33]

[Thermodynamics]

First law:

- (A) Energy conservation
- (B) Entropy
- (C) Enthalpy
- (D) Gibbs

Q11.

1M [no -ve]

[Thermodynamics]

First law:

*Enter your answer here***Q12.**

1M [-0.33]

[Computers and Programming]

Binary search complexity:

- (A) $O(n)$
- (B) $O(\log n)$
- (C) $O(n^2)$
- (D) $O(1)$

Q13.

1M [-0.33]

[Basic Electronics]

Diode used as:

- (A) Amplifier
- (B) Rectifier
- (C) Oscillator
- (D) None

Q14.

1M [no -ve]

[Fluid Mechanics]

Bernoulli assumes:

Enter your answer here

Q15.

1M [-0.33]

[Basic Electronics]

Diode used as:

- (A) Amplifier
- (B) Rectifier
- (C) Oscillator
- (D) None

Q16.

1M [-0.33]

[Engineering Mathematics]

If A is 2x2 with trace 5 and det 6, eigenvalues:

- (A) 2 and 3
- (B) 1 and 4
- (C) 3 and 3
- (D) 2 and 4

Q17.

1M [no -ve]

[Fluid Mechanics]

Bernoulli assumes:

*Enter your answer here***Q18.**

1M [-0.33]

[Environmental Science]

PM2.5 particle size:

- (A) $<2.5\mu\text{m}$
- (B) $<5\mu\text{m}$
- (C) $<10\mu\text{m}$
- (D) $<1\mu\text{m}$

Q19.

1M [-0.33]

[Computers and Programming]

Binary search complexity:

- (A) $O(n)$
- (B) $O(\log n)$
- (C) $O(n^2)$
- (D) $O(1)$

Q20.

1M [-0.33]

[Fluid Mechanics]

Bernoulli assumes:

- (A) Viscous
- (B) Inviscid
- (C) Compressible
- (D) Turbulent

Q21.

1M [no -ve]

[Basic Electrical]

V =:

*Enter your answer here***Q22.**

1M [-0.33]

[Engineering Mathematics]

If A is 2x2 with trace 5 and det 6, eigenvalues:

- (A) 2 and 3
- (B) 1 and 4
- (C) 3 and 3
- (D) 2 and 4

Q23.

1M [-0.33]

[Engineering Mathematics]

If A is 2x2 with trace 5 and det 6, eigenvalues:

- (A) 2 and 3
- (B) 1 and 4
- (C) 3 and 3
- (D) 2 and 4

Q24.

1M [-0.33]

[Material Science]

Thermoplastics:

- (A) Thermoset
- (B) Remoldable
- (C) Crosslinked
- (D) None

Q25.

1M [-0.33]

[Engineering Mathematics]

If A is 2x2 with trace 5 and det 6, eigenvalues:

- (A) 2 and 3
- (B) 1 and 4
- (C) 3 and 3
- (D) 2 and 4

Q26.

1M [no -ve]

[Fluid Mechanics]

Bernoulli assumes:

*Enter your answer here***Q27.**

1M [-0.33]

*[General Aptitude]*If $x^2=16$, $x =$:

- (A) 4
- (B) -4
- (C) ± 4
- (D) 0

Q28.

1M [-0.33]

[Basic Electronics]

Diode used as:

- (A) Amplifier
- (B) Rectifier
- (C) Oscillator
- (D) None

Q29.

1M [-0.33]

[Thermodynamics]

First law:

- (A) Energy conservation
- (B) Entropy
- (C) Enthalpy
- (D) Gibbs

Q30.

1M [no -ve]

[Thermodynamics]

First law:

*Enter your answer here***Q31.**

2M [no -ve]

[Environmental Science]

PM2.5 particle size:

*Enter your answer here***Q32.**

2M [no -ve]

[Fluid Mechanics]

Bernoulli assumes:

*Enter your answer here***Q33.**

2M [no -ve]

[Computers and Programming]

Binary search complexity:

*Enter your answer here***Q34.**

2M [-0.66]

[Environmental Science]

PM2.5 particle size:

- (A) <2.5; ~~µm~~
- (B) <5; ~~µm~~
- (C) <10; ~~µm~~
- (D) <1; ~~µm~~

Q35.

2M [-0.66]

[Environmental Science]

PM2.5 particle size:

- (A) $<2.5\mu\text{m}$
- (B) $<5\mu\text{m}$
- (C) $<10\mu\text{m}$
- (D) $<1\mu\text{m}$

Q36.

2M [-0.66]

[Thermodynamics]

First law:

- (A) Energy conservation
- (B) Entropy
- (C) Enthalpy
- (D) Gibbs

Q37.

2M [no -ve]

[Solid Mechanics]

Tensile stress formula:

*Enter your answer here***Q38.**

2M [-0.66]

[Solid Mechanics]

Tensile stress formula:

- (A) F/A
- (B) $F \cdot A$
- (C) F/A^2
- (D) A/F

Q39.

2M [-0.66]

[Material Science]

Thermoplastics:

- (A) Thermoset
- (B) Remoldable
- (C) Crosslinked
- (D) None

Q40.

2M [-0.66]

[Fluid Mechanics]

Bernoulli assumes:

- (A) Viscous
- (B) Inviscid
- (C) Compressible
- (D) Turbulent

Q41.

2M [no -ve]

[Material Science]

Thermoplastics:

*Enter your answer here***Q42.**

2M [-0.66]

*[General Aptitude]*If $x^2=16$, $x =$:

- (A) 4
- (B) -4
- (C) ± 4
- (D) 0

Q43.

2M [no -ve]

[Material Science]

Thermoplastics:

*Enter your answer here***Q44.**

2M [-0.66]

[Thermodynamics]

First law:

- (A) Energy conservation
- (B) Entropy
- (C) Enthalpy
- (D) Gibbs

Q45.

2M [-0.66]

[Basic Electronics]

Diode used as:

- (A) Amplifier
- (B) Rectifier
- (C) Oscillator
- (D) None

Q46.

2M [-0.66]

[Thermodynamics]

First law:

- (A) Energy conservation
- (B) Entropy
- (C) Enthalpy
- (D) Gibbs

Q47.

2M [-0.66]

[Material Science]

Thermoplastics:

- (A) Thermoset
- (B) Remoldable
- (C) Crosslinked
- (D) None

Q48.

2M [-0.66]

[Computers and Programming]

Binary search complexity:

- (A) $O(n)$
- (B) $O(\log n)$
- (C) $O(n^2)$
- (D) $O(1)$

Q49.

2M [no -ve]

[Basic Electrical]

V =:

- (A) IR
- (B) I/R
- (C) R/I
- (D) I^2R

Q50.

2M [-0.66]

[Solid Mechanics]

Tensile stress formula:

- (A) F/A
- (B) $F \cdot A$
- (C) F/A^2
- (D) A/F

Q51.

2M [no -ve]

[Computers and Programming]

Binary search complexity:

*Enter your answer here***Q52.**

2M [no -ve]

[Thermodynamics]

First law:

*Enter your answer here***Q53.**

2M [-0.66]

[Fluid Mechanics]

Bernoulli assumes:

- (A) Viscous
- (B) Inviscid
- (C) Compressible
- (D) Turbulent

Q54.

2M [no -ve]

[Basic Electrical]

V =:

Enter your answer here

Q55.

2M [no -ve]

[Environmental Science]

PM2.5 particle size:

*Enter your answer here***Q56.**

2M [-0.66]

[Fluid Mechanics]

Bernoulli assumes:

- (A) Viscous
- (B) Inviscid
- (C) Compressible
- (D) Turbulent

Q57.

2M [no -ve]

[Fluid Mechanics]

Bernoulli assumes:

- (A) Viscous
- (B) Inviscid
- (C) Compressible
- (D) Turbulent

Q58.

2M [no -ve]

[Material Science]

Thermoplastics:

*Enter your answer here***Q59.**

2M [-0.66]

[Basic Electrical]

V =:

- (A) IR
- (B) I/R
- (C) R/I
- (D) I^2R

Q60.

2M [-0.66]

[Computers and Programming]

Binary search complexity:

- (A) $O(n)$
- (B) $O(\log n)$
- (C) $O(n^2)$
- (D) $O(1)$

Q61.

2M [no -ve]

[Thermodynamics]

First law:

- (A) Energy conservation
- (B) Entropy
- (C) Enthalpy
- (D) Gibbs

Q62.

2M [-0.66]

[Material Science]

Thermoplastics:

- (A) Thermoset
- (B) Remoldable
- (C) Crosslinked
- (D) None

Q63.

2M [no -ve]

[Environmental Science]

PM2.5 particle size:

*Enter your answer here***Q64.**

2M [-0.66]

[Computers and Programming]

Binary search complexity:

- (A) $O(n)$
- (B) $O(\log n)$
- (C) $O(n^2)$
- (D) $O(1)$

Q65.

2M [-0.66]

[Fluid Mechanics]

Bernoulli assumes:

- (A) Viscous
- (B) Inviscid
- (C) Compressible
- (D) Turbulent